

Human Activity Recognition from Kinect Captured Data Using Stick Model

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Abstract. In this paper authors have presented a method to recognize basic human activities such as sitting, walking, laying, and standing in real time using simple features to accomplish a bigger goal of developing an elderly people health monitoring system using Kinect. We have used the skeleton joint positions obtained from the software development kit (SDK) of Microsoft as the input for the system. We have evaluated our proposed system against our own data set as well as on a subset of the MSR 3Ddaily activity data set and observed that our proposed method out performs state-of-the-art methods.

Keywords: Human activity, Human action, Kinect, Skeleton, Activity recognition.

1 Introduction

Now a days there is a great demand for elderly people health monitoring systems as elderly people are gradually increasing since last decade [1]. A demographic revolution is underway throughout the world. Today, world-wide, there are around 600 million persons aged 60 years and over; this total will double by 2025 and will reach virtually two billion by 2050 - the vast majority of them in the developing world. According to world health organization (WHO), the count of elderly people may reach to 2000 million by the year 2050 [2]. The statistics of elderly people from the year 2002 to 2050 predicted by WHO is shown in Fig. 1 [2]. Therefore, proper caring is very much needed to monitor their health by identifying their daily activities. A lot of systems were developed for monitoring the activities of people. Broadly all the systems are classified in to two categories, namely (i) ubiquitous sensor based and (ii) computer vision based approaches [3] [4]. But the problem with ubiquitous sensors is obtrusive and some sensors are invasive, so elderly people shows no or less interest in using them. Therefore, some researchers developed mobile phone based activity recognition. But the main problem comes with if the user can forget to carry mobile phone for some activities like going to bed, watching TV in drawing room, toileting etc. In such cases computer vision based approaches proved to be best one. The comprehensive survey on the human activity recognition in [3] concludes the requirement of depth sensor to make a robust computer vision based system for

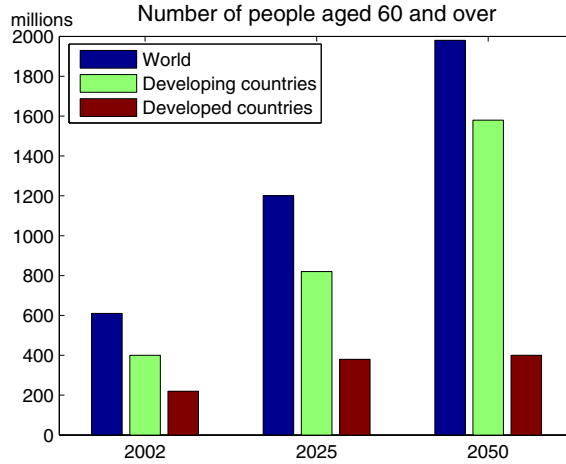


Fig. 1. Statistics of elderly people

human activity recognition. With the arrival of Kinect from Microsoft creates a new way of research in computer vision by mounting RGB camera with low cost depth sensor. Hence, we are using Kinect to monitor the activities in the bed room when the person may not carry the mobile phone. However, privacy is the major concern for monitoring bed room activities using computer vision based approach. But the advantage of deploying a Kinect based system is the availability of skeleton joint positions by using any stick model like Microsoft SDK so that the concern of privacy does not arise. We are using the floor map with the basic activities performed by the person in the bedroom recognized from the Kinect data to recognize the activities like going to toilet, coming back from toilet, going to bed, wake up from bed. But in this paper we are going to describe only our proposed method for activity recognition using the stick model obtained from the Kinect.

Rest of the paper is organized as follows: Following section presents the state-of-the-art works carried out in recognizing the activities. The data sets used for carrying out the experiments are explained in the section 3. The details of human activity recognition systems developed using different approaches is presented in the Section 4. The details of support vector machines for classifying the human activities is given in Section 5. Section 6 discuss the results of different approaches. Final section leads to summary and conclusions of the paper.

2 Prior Arts

Home activity monitoring is an interesting research topic for a long period. One such work in recent past can be found in [5]. Similarly research on human body model estimation using voxel data was started long back in the early of this century as [6]. But the onset of Kinect, the Microsoft gaming platform, [7] facilitate the access of 3-D data at a lower cost. Kinect was initially used for gesture

recognition to make the user feeling more comfortable while playing games [8]. Kinect can sense the Red-Green-Blue (RGB) color value of the pixels as well as the depth (D) value using an infra red sensor within the device. Software tool kits (SDK) and some open source codes are also available to obtain a stick model of skeleton points from the RGB-D data. Human activity recognition problems are solved by taking the RGB-D or the skeleton as input. In this work we focused on monitoring the human activities by using only skeleton information obtained from Kinect. The details of some of the existing works on human activity monitoring using skeleton information is given below.

In [9], on-board mobile robot is used and position and velocity of robot is used for predicting the human motion and position relative to robot. Spine is chosen as representative point. For estimating the relative position Kalman filter is used. Circular path and zigzag motion of human is considered for experimentation.

In [10], novel features such as local occupancy pattern (LOP) feature was used based on the depth data and the estimated 3D joint positions. In addition, new temporal pattern representation called Fourier Temporal Pyramid is used to represent the temporal structure of an individual joint in an action. The features used are robust to noise, invariant to translational and temporal misalignment, and capable of characterizing both the human motion and the human object interactions. An action-let ensemble model is learnt to represent each action and to capture the intra-class variance. An actionlet is a particular conjunction of features for a subset of the joints, indicating a structure of the features. As there are an enormous number of possible actionlets, novel data mining solution to discover discriminative actionlets. Discriminative weights are learnt by kernel method.

In [11], only 10 joints of skeleton such as Head, ShoulderCenter, Spine, HipCenter, HipLeft, HipRight, KneeLeft, KneeRight, AnkleLeft, AnkleRight are considered and 4 types of features are extracted for human posture recognition in the context of a health monitoring framework. 7 different experiments were carried out with the coordinates and different angles formed within these ten joints. With and without scaling the features is carried out. Support vector machine (SVM) was used for classifying the activities such as standing, sitting, bending and laying.

In [12], eigen joints features such as 3D position differences of joints to characterize action information including posture feature fcc, motion feature fcp, and offset feature fci in each frame and then concatenated the three features. Naive-Bayes-Nearest-Neighbor (NBNN) classifier is used for multi-class action classification.

In [13], Sequence of the Most Informative Joints (SMIJ) is used. Different sets of joints reveal discriminative information about the underlying structure of the action. At each time instant, automatically selects a few skeletal joints that are deemed to be the most informative for performing the current action. The selection of joints is based on highly interpretable measures such as the mean or variance of joint angles, maximum angular velocity of joints, etc. Histograms of Most Informative Joints (HMIJ), Histogram-of- MotionWords (HMW) and

Linear Dynamical System Parameters (LDSP), are used to demonstrate the power of the SMIJ features in terms of discriminability and interpretability for human action recognition. the quality of different features are evaluated using 1-nearest neighbor (1-NN) and support vector machine (SVM). Levenshtein distance is used for classification based on SMIJ. 2 distance is used for classification based on histogram feature representations HMIJ and HMW. Martin distance is used as a metric between dynamical systems for classification based on LDSP. One-vs-one classification scheme with Gaussian kernel is used.

In [14], histograms of 3D joint locations (HOJ3D) from 12 informative joints are used for compact representation of postures. The 12 joints includes head, L/ R elbow, L/ R hands, L/ R knee, L/ R feet, hip center and L/ R hip. Hip center was taken as the center of the reference coordinate system, and define the x-direction according to L/ R hip. The rest 9 joints are used to compute the 3D spatial histogram. Linear discriminant analysis (LDA) is performed to extract the dominant features. 3D skeletal joint locations are extracted from Kinect depth maps using Shotton method. Using LDA, the computed HOJ3D from the action depth sequences are reprojected. Later they clustered into k posture visual words, which represent the prototypical poses of actions. Discrete Hidden Markov models (HMMs) are used to model the temporal evolutions of visual words. It demonstrates significant view invariance on 3D action data set based on the design of spherical coordinate system and the robust 3D skeleton estimation from Kinect.

3 Activity Database

We evaluated the performance of our proposed method and state of the art methods on two different human activity data sets of 3D skeleton data. One data set is our own data set and second data set is standard MSR Daily Activity 3D data set. The description of each data set is given below.

Data set #1: The data set used for activity recognition is collected using Microsoft Kinect. The Kinect camera is fixed on the wall and recorded the activities. The sequence of different activities such as sitting, walking, laying and standing are recorded by covering all possible combinations or variations. The data is collected from 10 subjects consists of 5 male and 5 female. The subjects are within the age group of 23-40. For each subject we collected 2 repetitions of each action. Each repetition is about 90 sec duration, yielding a total of 120 min ($90\text{sec} \times 2\text{rep} \times 4\text{activities} \times 10\text{subjects}$) of data. The frame rate of Kinect sensor of skeleton data is 30. Therefore, total number of frames for each activity from all subjects obtained is $30 \times 60 \times 30 = 54000$ frames. The skeleton data collected for each activity from each subject is dumped into text file which contains the basic 20 skeleton joint positions (HipCenter, Spine ShoulderCenter, Head, ShoulderLeft, ElbowLeft, WristLeft, HandLeft, ShoulderRight, ElbowRight, WristRight, HandRight, HipLeft, KneeLeft, AnkleLeft, FootLeft, HipRight, KneeRight, AnkleRight, FootRight) with its X, Y and Z co-ordinates.

Data set #2: The methods used in this work for identifying the human activity is also tested on the standard MSR Daily Activity 3D data set consisting of the skeleton data obtained from a depth sensor similar to the Microsoft Kinect with 15 Hz. MSR DailyActivity3Ddataset is a daily activity data set captured by a Kinect device. There are 16 activity types: drink, eat, read book, call cellphone, write on a paper, use laptop, use vacuum cleaner, cheer up, sit still, toss paper, play game, lay down on sofa, walk, play guitar, stand up, sit down. If possible, each subject performs an activity in two different poses: sitting on sofa and standing. The total number of the activity samples is 320. Out of 16 activities we selected the subset of 4 activities such as sit still, lay down on sofa, walk and stand up(same activities which was mentioned earlier in Data set #1) for testing our methods.

4 Proposed Method

In the proposed method we have used the skeleton joints obtained from Kinect as the input. The Microsoft SDK we have used usually returns 20 skeleton joints with their x, y and z coordinates. It was reported in the literature [13] that all the joints are not required for activity recognition. They have reported 6 major joints to be most informative. So we have analyzed the skeleton joint points and observed that most of the joint points are very noisy irrespective of the sequence and some joints are not much affected from the noise. As per our analysis, Head, Shoulder- Center, Shoulder-Left and Shoulder-Right are the most reliable joints. Therefore, in this work we carried out the experimentation using these 4 joints. In this work we accomplish the task of human activity recognition in three steps namely (i) Feature extraction phase,(ii) Training phase, and (iii) Testing phase.

4.1 Feature Extraction

In this work we have explored three different features for recognizing human activities. This phase is the core part of any classification system as the performance of the system in terms of accuracy largely depends on set of features used. The details of the feature extraction methods are given below.

PCA Based Approach. In this method, simple X, Y, and Z co-ordinates of the skeleton joint positions of Head, ShoulderCenter, ShoulderLeft and ShoulderRight are considered for every 30 frames and then we applied principle component analysis (PCA). The steps involved in feature extraction is given below.

1. In this method, the X,Y, and Z coordinates of the corresponding 4 joints i.e., Head, ShoulderCenter, ShoulderLeft and ShoulderRight are extracted for every 30 frames.

2. The extracted points are arranged in the form of matrix $H_{12 \times 30}$ where rows represent the three coordinates of four joints and columns represent the frames.
3. The mean value for each row of matrix $H_{12 \times 30}$ is calculated i.e., $M_{12 \times 1}$ and subtracted the matrix M from H , resulting matrix $\tilde{H}_{12 \times 30}$.
4. Now the covariance of the matrix $\tilde{H}_{12 \times 30}$ is obtained by using $\tilde{H}_{12 \times 30} \times \tilde{H}_{12 \times 30}^T$. The resulting covariance matrix is $C_{12 \times 12}$, where each column represent the eigen vector. Among the 12 eigen vectors, top 7 eigen vectors are considered based on the top 7 eigen values. The resulting matrix now obtained is $P_{12 \times 7}$.
5. Original feature matrix $H_{12 \times 30}$ can be represented as $H_{12 \times 30} = \tilde{H}_{12 \times 30} + P_{12 \times 7} \times B_{7 \times 30}$, where $B_{7 \times 30}$ is the weight matrix.
6. The weight matrix is now obtained as $B_{7 \times 30} = P_{7 \times 12}^{-1} (H_{12 \times 30} - \tilde{H}_{12 \times 30})$
7. Now concatenate all the columns of $B_{7 \times 30}$ into single vector f_{PCA} forming 210 dimensional feature vector.
8. Follow steps 1 to 7 for each activity from each person data.

Statistical Features. From the collected data, we have analyzed that there is a lot of variation exists between the mean values of the X, Y and Z coordinates of the above mentioned four joints for each activity. The similar phenomenon is also observed between the difference of maximum and minimum values of X, Y and Z coordinates of the four joints for each activity. Hence in this work we have explored mean values and difference between maximum and minimum values of the joint positions of the four joints as features for identification of activities such as sitting, walking, laying and standing. Let F_{mean} be the feature vector which are the mean values of the 3 coordinates of 4 joints extracted for every 30 frames. Therefore the feature vector F_{mean} is represented by 12 features. Let $F_{max-min}$ be the feature vector which are the values of difference between maximum and minimum values of the X, Y and Z coordinates of 4 joints extracted for every 30 frames. Therefore the feature vector $F_{max-min}$ is represented by 12 features. An example of discrimination of X, Y and Z co-ordinates of mean feature vector and the difference of maximum and minimum feature vector for the 4 joints for the corresponding 4 activities is shown in Fig. 2.

In this study, three human activity recognition systems(HARS) are developed which are summarized as follows:

1. HARS-1: Human activity recognition system using only mean values of the joint positions as features.
2. HARS-2: Human activity recognition system using only difference between maximum and minimum values of the joint positions as features.
3. HARS-3: Human activity recognition system using combination of mean values and difference between maximum and minimum values of the joint positions as features.

The experimentation is carried out using only mean values feature vector F_{mean} , only difference of maximum and minimum values feature vector $F_{max-min}$ and

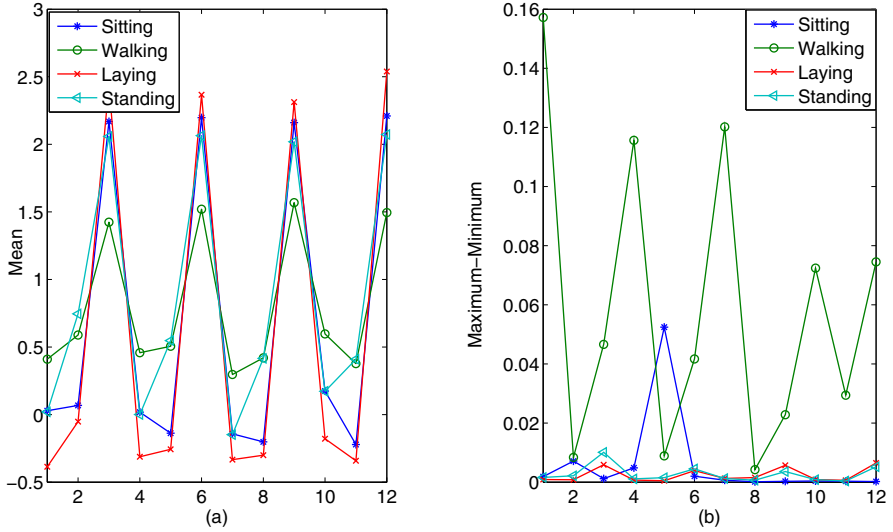


Fig. 2. Representation of feature vector (a) mean and (b) difference of maximum and minimum of the four activities sitting, walking, laying and standing

combination of both $F=[F_{mean}, F_{max-min}]$. It is observed the concatenated feature vector outperformed compared to individual feature vectors. This can be verified from the results given in the Section 6.

4.2 Training

For developing the efficient human activity recognition system, proper training need to be carried out using machine learning. In this work 5 fold cross validation is used, where 4 folds used for training and 1 fold for testing. The sequence of steps followed in training phase are given below.

1. Let A be a set of m activities ($A = a_1, a_2, \dots, a_m$) available to us.
2. Let S be a set of annotated skeleton data available.
3. Let $\theta_k \in S$ be the set of data used for training phase, where θ_k is a subset of data of set S and θ_k is 80% of S .
4. Now use feature vectors extracted in the feature extraction phase and activity pairs as input to machine learning from the training data θ_k to generate the models.

4.3 Testing

For testing the human activity recognition systems developed using different features, we used 1 fold of data out of 5 folds. Testing is carried by using the

models developed in training phase. The sequence of steps followed in testing phase are given below.

1. Let $\beta_k \in S$ be the set of data used for testing phase, where β_k is a subset of data of set S and β_k is 20% of S .
2. The data in the test set is not present in the train set i.e., $\theta_k \cap \beta_k = \phi$.
3. The same steps are followed for extraction of features from test set β_k of different methods which was mentioned in the previous subsection.
4. Now in the testing phase we provide only the features (f) as input to the system.
5. Now system predicts the activities based on the learning models developed in the training phase.
6. For each method experiment is run for 5 times such as out of 5 folds of data, each time 1 fold is used for testing and remaining 4 fold used for training.
7. Now the performance accuracy in terms of percentage is computed for each run in test phase as follows:

$$\%Accuracy = \frac{A_c}{A_t} \times 100$$

where A_c is number of activities correctly classified out of total activities A_t

8. The overall average performance accuracy of the test data is calculated by taking the mean of all accuracies obtained in the 5 runs carried out in test phase.

5 Support Vector Machines

In this work, Support Vector Machines (SVM) are explored as a machine learning tool to discriminate the human activities. SVM classification is an example of supervised learning. SVMs are useful due to their wide applicability for classification tasks in many signal processing applications. A classification task usually involves training and testing data which consist of some data instances. In the training set, each instance contains one target class label and many attributes. The main goal of SVM for classification problem is to produce a model which predicts target class label of data instances in the testing set, given only the attributes. The SVM models for different human activities were developed as one against-rest principle. The SVM model for the specific activity was developed, by using feature vectors derived from the desired human activity clues as positive examples and the feature vectors derived from the other human activities as negative examples. Radial basis function (RBF) kernel, unlike linear kernel, is used in this work to map the data points to higher dimensional space as it can handle the case where the relation between the class labels and attributes is nonlinear. The intuition to use RBF kernel function is due to its universal approximation properties. Also, it offers good generalization as well as good performance in solving practical problems [15]. The basic architecture of human activity classification system using SVMs with above mentioned features is shown in Fig. 3.

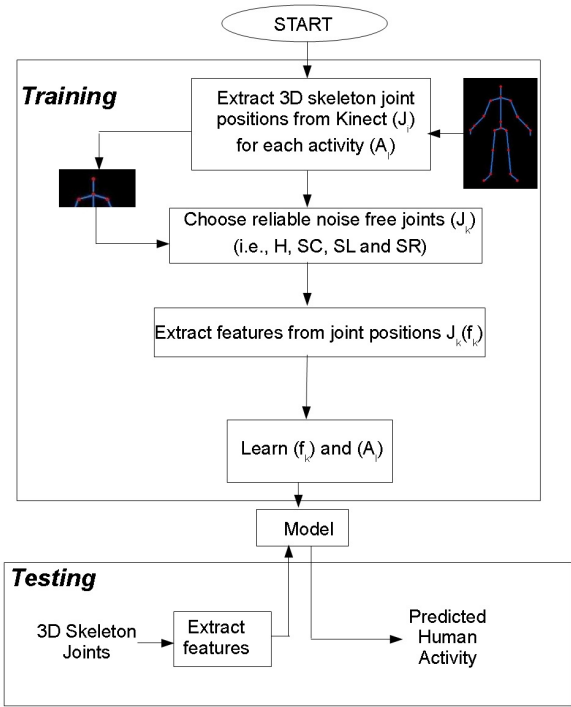


Fig. 3. Architecture of activity recognition system(H:Head, SC: Shoulder Center, SL:Shoulder Left and SR:Shoulder Right

6 Experimental Results

We have tested all the activity recognition methods which are mentioned above on Data set #1 and Data set #2. The features extracted from above mentioned methods are used as input for SVM and activities as labels to SVM. In this work, we used 5 fold cross validation and then the average performance is computed. The performance accuracy of the different methods mentioned above is given in Table 1. Columns 1 and 2 of Table 1 indicates the data sets used for evaluation of the methods (rows of Table 1). The values in Table 1 indicate the average performance accuracy of different methods. The performance of activity recognition using PCA features observed to be poor. The poor performance is mainly due to dependency on the distance and their absolute values. The performance of PCA based approach is improved by normalizing in between -1 and 1 and thereby making it uniform and independence of distance. Normalization reduce the intr-class variation under different test sets. The drastic improvement in the performance of activity recognition for without and with normalization of PCA features can be observed from the rows 1 and 2 of Table 1. From Table 1, it

is observed that the performance accuracy of the method using only the mean values of the X, Y, and Z co-ordinates of skeleton data extracted for every 30 frames performed better compared to other individual methods. It is observed that for data set #2, the average performance of human activities using difference between maximum and minimum values seems to be poor compared to data set #1. This is mainly due to more randomness of data present in data set #2. But the combination of features such as mean values, and difference of maximum and minimum values outperformed compared to individual features for both data sets. The performance accuracy of human activities of different methods for data sets #1 and #2 is also plotted in Fig. 4.

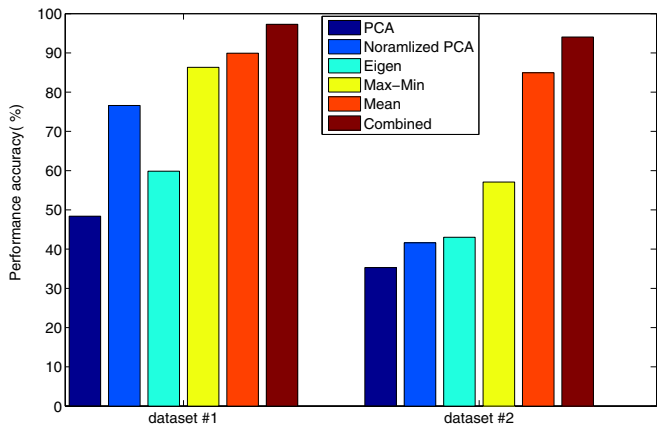


Fig. 4. Performance plot of human activities of different methods

Table 1. Performance of the different methods for identification of activities)

Sl. No	Features	Average Classification Performance (%)	
		Our dataset	MSR 3Ddaily activity dataset
1	PCA	48.40	35.30
2	Normalized PCA	76.60	41.62
3	EigenJoints	59.86	43.03
4	Maximum-Minimum(Range)	86.32	57.10
5	Mean	89.93	84.95
6	Combination 4 and 5 (Range and Mean)	97.29	94.06

7 Summary and Conclusions

In this paper we have presented a skeleton joint based method for activity recognition where we have used four major joint points that reduces the possibility of error due to noise as well as it reduces the over all time complexity of the system. We have also explored two different features for recognizing the activity. We have compared our features against eigen joint based approach and find that

our method out performs that. Our system can recognize these basic activities up to 97.29% accuracy which is much better than the state of the art. Our system is not working in some sequences as our system is not using any noise cleaning method. We are working on incorporating any suitable de-noising method. We are now working to integrate it with our live system so that it can be deployed in homes to monitor the activities of the elderly people.

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